

7.Unification and Resolution – Exercises with Solutions

Exercise 1: Sentence Matcher

```
% Facts
sentence(the_cat_sits).
sentence(the_dog_runs).
```

```
% Query
% ?- sentence(X).
```

```
Output:
22 ?- consult(['c:/Users/HDCO422166/Downloads/facts.pl']).
true.
```

```
23 ?- sentence(X).
X = the_cat_sits .
```

Exercise 2: Compound Term Unification

```
% Facts
likes(john, food(pizza, spicy)).
likes(mary, food(pizza, sweet)).
```

```
% Query
% ?- likes(john, food(pizza, T)).
```

```
Output:
24 ?- consult(['c:/Users/HDCO422166/Downloads/food.pl']).
true.
```

```
25 ?- likes(john, food(pizza, T)).
T = spicy .
```

Exercise 3: Conflict Resolution

```
% Facts
rule1(climate_change).
rule2(climate_change).

% Rule
conflict(X) :- rule1(X), rule2(X).

% Query
% ?- conflict(climate_change).
```

Output:

```
26 ?- consult(['c:/Users/HDCO422166/Downloads/weather.pl']).
true.
```

```
27 ?- conflict(climate_change).
true.
```